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Effects of photobiomodulation therapy combined with static magnetic field on pain and function in patients with lateral epicondylitis: a multicentre, randomised, placebo-controlled trial

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ABSTRACT

Introduction Photobiomodulation therapy (PBMT), particularly when combined with a static magnetic field (PBMT-sMF), is a promising non-pharmacological approach for managing musculoskeletal disorders. However, high-quality evidence for its efficacy in lateral epicondylitis remains limited.

Objectives The study aims to investigate the effectiveness of PBMT-sMF vs placebo in reducing pain, improving function and modulating inflammatory markers in individuals with lateral epicondylitis.

Design Multicentre, randomised, triple-blinded, placebo-controlled trial.

Setting Three outpatient physiotherapy clinics in Brazil.

Participants 50 adults (18–50 years) with unilateral lateral epicondylitis and baseline pain ≥50 on the visual analogue scale (VAS).

Interventions Participants received either active PBMT-sMF (n=25) or placebo (n=25), 2 times per week for 3 weeks. PBMT-sMF involved multi-wavelength irradiation at 4 epicondyle sites (60 s; 27.1 J/site). The placebo group underwent the same procedure without active irradiation.

Primary and secondary outcome measures The primary outcome was degree of pain rating (VAS). Secondary outcomes included forearm disability (Patient-Rated Tennis Elbow Evaluation, PRTEE), grip strength, serum tumour necrosis factor-alpha (TNF-α) levels and treatment satisfaction. Assessments were conducted at baseline, post-treatment (3 weeks) and at 4-week follow-up.

Results PBMT-sMF yielded a higher responder rate (defined as the proportion of participants achieving at least a 30% reduction in pain intensity relative to baseline) than placebo (72% vs 40%, p=0.045), with a clinically and statistically significant between-group difference. Compared with placebo, the PBMT-sMF group showed significantly greater reductions in pain intensity both at the end of treatment (51.4±19.8 vs 36.9±22.6; p=0.0223) and at follow-up (37.4±24.1 vs 20.3±21.2; p=0.0049). TNF-α levels also decreased significantly in the PBMT-sMF group compared with placebo at both time points (p<0.0001 and p=0.0019, respectively). No significant between-group differences were observed for PRTEE scores, grip strength or treatment satisfaction. No major adverse events were reported.

Conclusions PBMT-sMF significantly reduced pain intensity and TNF-α levels, suggesting an anti-inflammatory mechanism. Although functional outcomes were not improved, PBMT-sMF may be a valuable short-term, non-invasive option for lateral epicondylitis pain management.

Trial registration number NCT04829734 on ClinicalTrials.gov

STRENGTHS AND LIMITATIONS OF THIS STUDY

⇒ Short-term assessment only: A key limitation of the present trial is that it focused exclusively on the short-term effects of photobiomodulation therapy combined with a static magnetic field, without evaluating long-term outcomes.
⇒ Robust methodological design: The study was conducted across multiple centres and implemented strict blinding procedures for participants, therapists and outcome assessors.
⇒ Statistical rigour: The sample size was determined through a priori calculation to ensure adequate power for detecting clinically significant differences between groups.
⇒ Protocol adherence and trial integrity: The trial was prospectively registered, followed the predefined protocol without deviations and included key methodological safeguards such as true randomisation, allocation concealment and analysis based on the intention-to-treat principle.
⇒ Control for confounding variables: The inclusion of a placebo group allowed for control of potential confounders such as the placebo effect, regression to the mean, therapist bias and co-interventions.

INTRODUCTION

Musculoskeletal disorders affecting the upper extremities are widespread in the general population, including cervical disc disease, rotator cuff injuries, carpal tunnel syndrome, osteoarthritis and epicondylitis.[1] These conditions can adversely affect quality of life and lead to disability, increased healthcare utilisation and reduced productivity.[2–4] Lateral epicondylitis, often called 'tennis elbow,' is among the most common of these disorders and a primary cause of elbow pain in adults.[5 6] It is frequently attributed to repetitive occupational or recreational tasks, particularly in racquet sports.[6 7] Although lateral epicondylitis is traditionally viewed as an overuse injury, some evidence indicates a degenerative process involving tendinosis, microtrauma and tears in the extensor carpi radialis brevis.[8] Most cases resolve within 6 months, but a subset of patients experiences symptoms for up to 2 years,[6 7] leading to considerable healthcare expenses and productivity losses.[7]

Non-surgical interventions, including oral medications, corticosteroid injections and physiotherapy, are commonly employed for lateral epicondylitis management.[9] Among physiotherapeutic approaches, photobiomodulation therapy (PBMT)—which uses laser and light-emitting diodes (LEDs) to interact with mitochondrial chromophores—has garnered attention for its ability to stimulate tissue repair, modulate inflammation and alleviate pain.[10] In some devices, PBMT is combined with a static magnetic field (PBMT-sMF) to further enhance these effects.[11 12] Studies have shown that PBMT may accelerate tendon repair by improving tissue remodelling, promoting collagen deposition and reducing inflammation.[13–15]

Despite its frequent clinical use and plausible biological rationale, however, few high-quality randomised controlled trials have evaluated PBMT for lateral epicondylitis. Existing evidence remains inconsistent, with some studies reporting no difference between PBMT and placebo,[16–18] and others demonstrating favourable outcomes.[19 20] A systematic review suggested that PBMT can reduce pain and disability in lateral epicondylitis, particularly when optimal parameters are employed.[21] Given the considerable burden that lateral epicondylitis places on both the healthcare system and society at large, and in light of the conflicting findings regarding PBMT treatment, further methodologically robust trials are needed to clarify its therapeutic potential, especially with well-defined parameters, on lateral epicondylitis. Therefore, this study aimed to investigate the effects of PBMT-sMF vs placebo on pain intensity, disability, grip strength, inflammatory marker levels and treatment satisfaction in individuals diagnosed with lateral epicondylitis.

MATERIALS AND METHODS

Design

This multicentre trial was designed as a superiority, parallel-group, randomised, placebo-controlled study with triple blinding (patients, therapists and outcome assessors). The protocol was prospectively registered on ClinicalTrials.gov (NCT04829734) and approved by the Research Ethics Committee of Nove de Julho University (#4.705.746). All procedures were conducted in accordance with the Declaration of Helsinki, and written informed consent was obtained from every participant. Participants and/or the public were not involved in the design, recruitment or conduction of this study; the main results were disseminated individually to participants by email. No protocol amendments were made during the study.

Setting and participants

Between April and September 2021, the study was conducted at 3 outpatient physiotherapy clinics in Brazil. Participants were recruited via flyers and print advertisements and were required to be seeking treatment for lateral epicondylitis. Inclusion criteria included a history of lateral epicondyle pain for at least 1 month; a minimum pain intensity of 50 on a 0–100 visual analogue scale (VAS); localised tenderness at the epicondyle and anterodistal region on palpation and at least 2 positive findings among 4 provocative tests (Maudsley's, Cozen's, Thomsen's and Mill's tests). Eligible individuals were aged 18 to 50, of any gender, and fluent in Portuguese.

Exclusion criteria were as follows: known clotting disorders (including haemophilia); chronic immune impairments; current or recent (<6 months) cancer treatment (including spinal tumours); type 1 diabetes; significant cardiovascular diseases (e.g., congestive heart failure, pacemaker use); active chronic pain conditions (such as chronic fatigue syndrome, fibromyalgia, endometriosis, inflammatory bowel disease, interstitial cystitis or diabetic neuropathic pain); neurological deficits including cervical radiculopathy; peripheral nerve disorders; rheumatoid arthritis; shoulder pathology; radial tunnel syndrome; previous surgery on the affected limb; congenital or acquired bony deformities in the ipsilateral upper extremity; bilateral epicondylosis; secondary orthopaedic conditions; recent initiation of opioid or corticosteroid/analgesic injections (within the last 6 months); local injections of corticosteroids or botulinum toxin (Botox) within the prior 30 days; recent treatments (such as chiropractic care or acupuncture in the last 30 days); any physical therapy on the affected upper extremity during the previous year; active infection, wound or trauma in the treatment area; contraindications or hypersensitivity to light therapy; severe mental health conditions (e.g., dementia or schizophrenia) or psychiatric hospitalisation within the past 2 years and pregnancy, lactation or plans to conceive during the study period.

Randomisation

The website *random.org* was used to generate a simple randomisation schedule (allocation ratio 1:1). The PBMT device used was previously programmed into active or placebo mode according to this randomisation schedule.

Both randomisation and device programming were performed by one of the study investigators not directly involved with the assessment and treatment of patients. Allocation concealment was achieved using sealed and opaque envelopes, consecutively numbered.

A simple randomisation sequence was generated using random.org with a 1:1 allocation ratio. The PBMT-sMF device was pre-programmed into either active or placebo mode according to this sequence. An investigator not involved in patient treatment or assessment handled the randomisation and device programming and allocation concealment was maintained by using consecutively numbered, sealed, opaque envelopes.

Blinding

Triple blinding was maintained throughout the study, ensuring that patients, therapists and outcome assessors remained unaware of the treatment allocations. The device, which produces no thermal effects,[22] emitted identical sounds and displayed the same information regardless of whether it was set to active or placebo mode.

Interventions

Both treatment groups underwent sessions 2 times per week over 3 consecutive weeks, with intervals of 72– 96 hours between sessions, for a total of 6 sessions. All treatments were administered using the MRM MR5 Prototype (Multi Radiance Medical, Solon, OH, USA). The interventions were specified as follows.

Active PBMT-sMF

The MRM MR5 Prototype includes a super-pulsed infrared laser (905 nm), 3 infrared LEDs (850 nm) and 3 red LEDs (640 nm). PBMT-sMF was administered to 4 regions of the epicondyle. Each site received a 60-s application, delivering an energy dose of 27.1 J per site, resulting in a total energy delivery of 108.3 J per treatment session. The parameters are detailed in table 1.

Placebo PBMT-sMF

The placebo PBMT-sMF treatment followed the same procedure as the active PBMT-sMF treatment. The same application sites and device were used; however, the 905 nm laser diodes, 875 nm LED diodes and static magnetic field were deactivated. Additionally, the power of the 640 nm LED diodes was reduced to 1 mW (mean power per diode) to maintain the visual appearance of red light without delivering a therapeutically effective dose. The total energy delivered per region was less than 1 J.

Table 1 Parameters for MRM MR5 prototype

Class	1M
Number of lasers	1 Super-pulsed infrared
Wavelength (nm)	905 (±1)
Frequency (Hz)	250
Peak power (W)	50
Average mean optical output (mW)	1.25
Power density (mW/cm^2)	2.84
Energy density (J/cm^2)	0.170
Dose (J)	0.075
Spot size of laser (cm^2)	0.44
Number of red LEDs	3 Red
Wavelength of red LEDs (nm)	640 (±10)
Frequency (Hz)	2
Average optical output (mW)—each	66.67
Power density (mW/cm^2)—each	74.08
Energy density (J/cm^2)—each	4.44
Dose (J)—each	4.00
Spot size of red LED (cm^2)—each	0.9
Number of infrared LEDs	3 Infrared
Wavelength of infrared LEDs (nm)	875 (±10)
Frequency (Hz)	16
Average optical output (mW)—each	83.33
Power density (mW/cm^2)—each	92.59
Energy density (J/cm^2)—each	5.55
Dose (J)—each	5.00
Spot size of LED (cm^2)—each	0.9
Magnetic field (mT)	35
Irradiation time per site (sec)	60
Total energy irradiated per site (J)	27.1
Number of sites irradiated per treatment	4
Total energy irradiated per treatment (J)	108.3
Aperture of the device (cm^2)	4
Application mode	Cluster probe held stationary in skin contact with a 90° angle and slight pressure

LED, light-emitting diode.

Outcomes

Primary outcome

The primary outcome was degree of pain rating at the end of the treatment period (3 weeks). Pain intensity was quantified using the 100 mm horizontal VAS, a validated tool in which patients mark their perceived pain along a continuum ranging from 0 (no pain) to 100 (the worst possible pain).

Secondary outcomes

A series of secondary outcomes were evaluated to capture additional dimensions of treatment efficacy and safety, both at the end of treatment (3 weeks) and at a follow-up (4 weeks post-treatment, equivalent to 1 month after treatment cessation):

- Degree of pain rating at 4 weeks post-treatment: Pain intensity was reassessed using the same VAS methodology at 4 weeks post-treatment.
- Forearm pain and disability: Pain and functional limitations associated with lateral epicondylitis were measured using the Patient-Rated Tennis Elbow Evaluation (PRTEE).[23] The PRTEE comprises 15 items divided into 2 subscales: a PAIN subscale contained 5 questions about pain intensity during specific activities and at rest (each item is rated from 0=no pain and 10=worst pain imaginable), and a FUNCTION subscale contained 10 questions about how much difficulty the patient experiences when performing various daily or work-related tasks (each item is rated from 0=no difficulty to 10=inability to perform activities). These subscales are combined to yield a total score ranging from 0 to 100, with higher scores indicating greater impairment.
- Grip strength: Functional improvement was also assessed by measuring grip strength using a digital dynamometer (Jamar Plus Digital Hand Dynamometer, Patterson Medical, Warrenville, IL, USA). For the measurement, the dynamometer was positioned on the second joint of the fingers to capture the force generated during a standardised grip test. Patients were instructed to maintain a maximum grip for 5 seconds during a period without pain, with both arms kept in a relaxed position to minimise external influence. Three consecutive measurements were taken, and the average value was used for analysis.[24]
- Serum tumour necrosis factor-alpha (TNF-α) levels: As an objective marker of inflammation, TNF-α levels were measured from serum samples obtained via venipuncture from the antecubital vein. After centrifugation, the serum was analysed using an ELISA in accordance with the manufacturer's protocol.
- Patient satisfaction with overall outcome: Treatment satisfaction was assessed using a 5-point Likert scale ranging from 1 ('very unsatisfied') to 5 ('very satisfied'), with the following response options: (1) very unsatisfied, (2) unsatisfied, (3) neutral (neither satisfied nor unsatisfied), (4) satisfied, and (5) very satisfied.
- Adverse events and co-interventions: Safety was monitored by systematically recording any adverse events throughout the study. Participants documented any side effects, discomfort or unexpected symptoms in a daily diary during both the treatment and follow-up phases. This diary also recorded any co-interventions, such as additional medications or therapies used to manage epicondylitis symptoms. These detailed records were reviewed to assess the overall safety profile and tolerability of the intervention.

Procedures

The study was conducted in 5 phases. (1) In the Pre-Procedure Activities phase, patients underwent initial screening via medical history and clinical examination to confirm eligibility. After signing the informed consent form and being randomised, participants entered a 1-week stabilisation phase during which they recorded their current pain management regimens in a daily diary and VAS pain ratings over the final 3 days. Patients were allowed to continue their usual pain management regimens throughout the study; however, these regimens were strictly limited to medications and non-pharmacological therapies they had already been using at the time of study entry. No new interventions were permitted. The specific treatments, medications and usage instructions were documented by the investigator at the beginning of the stabilisation phase and monitored daily via the Subject Diary. (2) In the Pre-Procedure Assessment phase, final eligibility was confirmed based on a VAS pain intensity of ≥50 during the 3 days preceding this phase. Baseline demographic and clinical data were then collected. (3) The Procedure Administration phase involved 6 treatment sessions (either active or placebo PBMT-sMF) over a 3-week period, with sessions spaced 3 to 4 days apart. (4) During the Procedure Administration Phase Measures, participants continued to complete their daily diaries, and primary and secondary outcomes were measured within 24 hours following the final session. (5) In the Post-Procedure Phase, during the week following the treatment phase, patients maintained their daily diaries. 4 weeks after completing the treatment phase, participants returned for a final follow-up evaluation, at which time all outcome measures and diary records were collected.

Sample size

The predefined criterion for individual treatment success was a ≥30% reduction in pain intensity from baseline to the study endpoint. This threshold is supported by the Initiative on Methods, Measurement and Pain Assessment in Clinical Trials (IMMPACT) recommendations, which propose that a 30% reduction in pain reflects a moderate clinically important improvement from the patient's perspective and is appropriate for interpreting meaningful change in chronic pain trials.[25] The proportion of patients classified as responders has gained prominence in the analysis, interpretation, publication and assessment of clinical trials on pain interventions. Notably, responder rates are easily interpretable and facilitate the comparison of outcomes across studies.[25]

Based on a pilot study conducted by our research group with 20 participants (10 per group), 60% of patients in the active treatment group and 20% in the placebo group met the individual success criterion. Using a 2-tailed test with an alpha level of 0.05 and a power of 0.8, the estimated required sample size was 25 participants per group.[26] Accordingly, we recruited a total of 50 patients with lateral epicondylitis, assigning 25 to each group.

Statistical analysis

Statistical analyses were conducted according to the intention-to-treat principle. The researcher responsible for the statistical analysis was blinded to the randomisation

and group allocation of participants. Data normality was assessed using the Shapiro–Wilk test. As the data followed a normal distribution, demographic variables were analysed using 2-tailed Student's t-tests and Fisher's exact test.

For the primary outcome, Fisher's exact test was used to compare the proportion of responders between groups at the endpoint. In a secondary analysis of the primary outcome, as well as for secondary continuous outcomes, one-way analysis of covariance (ANCOVA) was employed. In these models, the mean change from baseline was used as the dependent variable, the intervention group as the independent variable, and baseline values as covariates.

Regarding patient satisfaction, responses on the Likert scale were dichotomised into 2 categories: 'accept' (including 'satisfied' and 'very satisfied') and 'reject' (including 'very unsatisfied,' 'unsatisfied' and 'neutral'). Data were analysed using the chi-square test. Fisher's exact test was also applied to assess differences in the frequency of co-interventions.

All statistical tests were two-tailed, and a p-value of <0.05 was considered statistically significant. Results were presented as absolute values and percentage change.

RESULTS

50 patients diagnosed with lateral epicondylitis participated in the study. They were randomised into either the placebo (n=25) or the PBMT-sMF (n=25) group, with baseline demographic and clinical characteristics comparable between the groups (see table 2). All participants were recruited and completed the study procedures between April and September 2021, and each received the treatment as assigned (figure 1). No participants discontinued the study or were lost to follow-up.

According to the predefined criterion for individual treatment success, the proportion of responders was significantly higher in the PBMT-sMF group (72%) compared with the placebo group (40%), with a between-group difference of 32% (p=0.045). At baseline, both groups exhibited similar VAS pain scores. A significant reduction in pain intensity (VAS) over time was observed in both groups (p<0.05). In addition, in the secondary analysis for the primary outcome, a significant between-group difference in pain intensity was observed in favour of PBMT-sMF at both the end of treatment (p=0.0223) and the follow-up assessment (p=0.0049). Additionally, compared with the placebo group, PBMT-sMF showed a significantly greater percentage reduction in pain intensity at the end of treatment (p=0.0060) and at the follow-up assessment (p=0.0018) (figure 2).

A similar pattern emerged for TNF-α levels, with PBMT-sMF demonstrating a significant advantage over placebo in both absolute values at the end of treatment (p<0.0001) and at follow-up assessment (p=0.0019), as well as in the percentage change at the end of treatment (p=0.0003) and at follow-up assessment (p=0.0098) (figure 3).

Table 2 Demographic and clinical characteristics of the patients at baseline (n=50)

Variables	PBMT-sMF (n=25)	Placebo (n=25)
Gender		
Female	12 (48%)	13 (52%)
Male	13 (52%)	12 (48%)
Age (years)	34.6 (7.8)	35.1 (9.8)
Weight (kg)	75.7 (17.2)	76.5 (18.2)
Height (cm)	166 (5)	167 (8)
Dominant limb breakdown		
Right	25 (100%)	24 (96%)
Left	0 (0%)	1 (4%)
Affected limb breakdown		
Right	16 (64%)	18 (72%)
Left	9 (36%)	7 (28%)
Lateral epicondyle pain (VAS 0–100 mm)	79 (10)	76 (10)
Grip strength (kg)		
Injured	23.1 (8.2)	24.2 (12.0)
Not injured	24.7 (7.8)	27.5 (12.5)
TNF-α (pg/mL)	35.7 (5.0)	36.7 (7.0)
Forearm pain and disability (PRTEE 0–100)	58 (9)	59 (8)

PMBT-sMF, photobiomodulation therapy combined with static magnetic field; PRTEE, Patient-Rated Tennis Elbow Evaluation; TNF-α, tumour necrosis factor-alpha; VAS, visual analogue scale.

In contrast, no significant between-group differences were found in forearm pain and disability, as measured by the PRTEE, at either the end of treatment or the follow-up assessment (p>0.05). Likewise, there were no between-group differences in grip strength or patient satisfaction with overall outcome at any time point (p>0.05). Table 3 shows the mean and standard deviation (absolute values) for primary and secondary outcomes.

At both time points, the majority of participants in both groups reported satisfaction with the treatment. The proportion of patients classified as 'accept' (ie, satisfied or very satisfied) was higher in the PBMT-sMF group compared with the placebo group, both at the end of treatment and at follow-up. However, no statistically significant between-group differences were observed in patient satisfaction, either at the end of treatment (p=0.2661) or at follow-up (p=0.3226). Very few participants reported neutral or unsatisfactory responses, and none reported being 'not at all satisfied'. Full data are available in online supplemental table 1.

Finally, no significant between-group differences (p>0.05) were observed in patient-reported

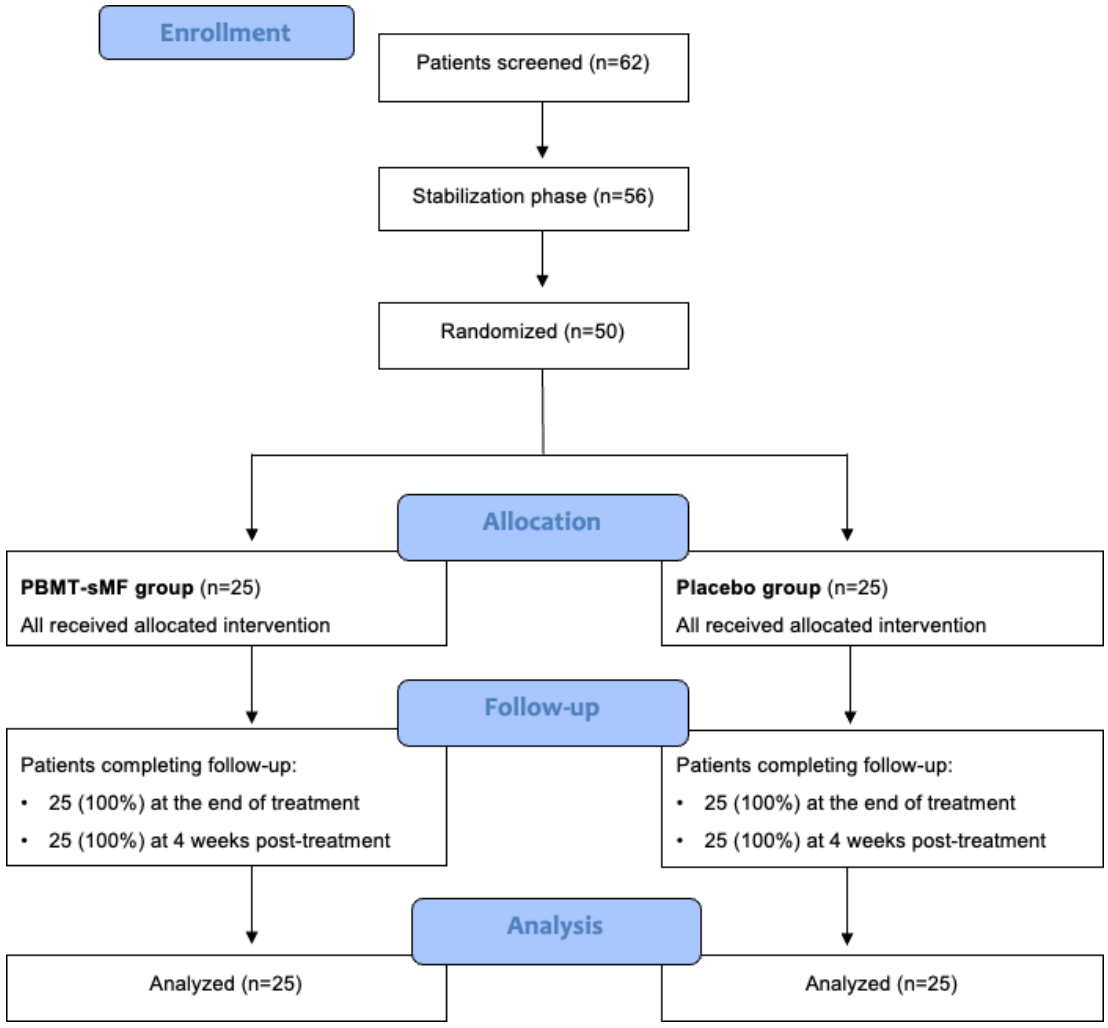


Figure 1 Flow diagram of the study. PMBT-sMF: photobiomodulation therapy combined with static magnetic field.

co-interventions (medications and other therapies) during the stabilisation, procedure and post-procedure phases (table 4). Reported medications were primarily pain-relieving drugs, such as non-steroidal anti-inflammatory drugs (NSAIDs) and analgesics.

No major adverse events were reported. However, 2 patients (8%) in the PBMT-sMF group experienced a strong 'biting' sensation and a heating sensation of the skin. In the placebo group, 3 patients (12%) reported pain, discomfort, increased pain or tingling. None of these events required intervention or led to participant withdrawal, and all adverse effects were fully resolved by the end of the procedure administration.

DISCUSSION

In this multicentre, triple-blinded, randomised, placebo-controlled trial, we investigated the effects of PBMT-sMF vs placebo on pain intensity, disability, grip strength, inflammatory marker levels, and treatment satisfaction in patients with lateral epicondylitis. Although both groups showed reductions in pain intensity and TNF-α levels, the PBMT-sMF group achieved significantly greater improvements at both the end of treatment and the follow-up assessments. No between-group differences were observed for the other outcomes.

Evidence regarding the effects of PBMT and PBMT-sMF on lateral epicondylitis remains scarce and conflicting. While some trials have found no positive effects of PBMT in these patients,[27 28] others have reported that PBMT outperforms placebo.[18–20] Notably, the two trials observing no significant benefits for PBMT both compared it to extracorporeal shockwave therapy (ESWT) rather than placebo.[27 28] Although some outcomes—such as pain intensity (VAS) and disability (PRTEE)—were comparable to those in our study, ESWT proved more effective than PBMT for reducing both pain and disability.[27] By contrast, Celik *et al*[28] found ESWT to be superior to PBMT only for grip strength, which aligns with our own results showing no between-group difference in this outcome. It is worth noting that we did not include an exercise protocol in our trial, and the PBMT parameters employed in those studies differed from ours, with substantially lower doses (0.25–2.4 J), as well as varied treatment frequencies (ranging from daily sessions to 3 times per week) and a total of 12–15 irradiation sessions.

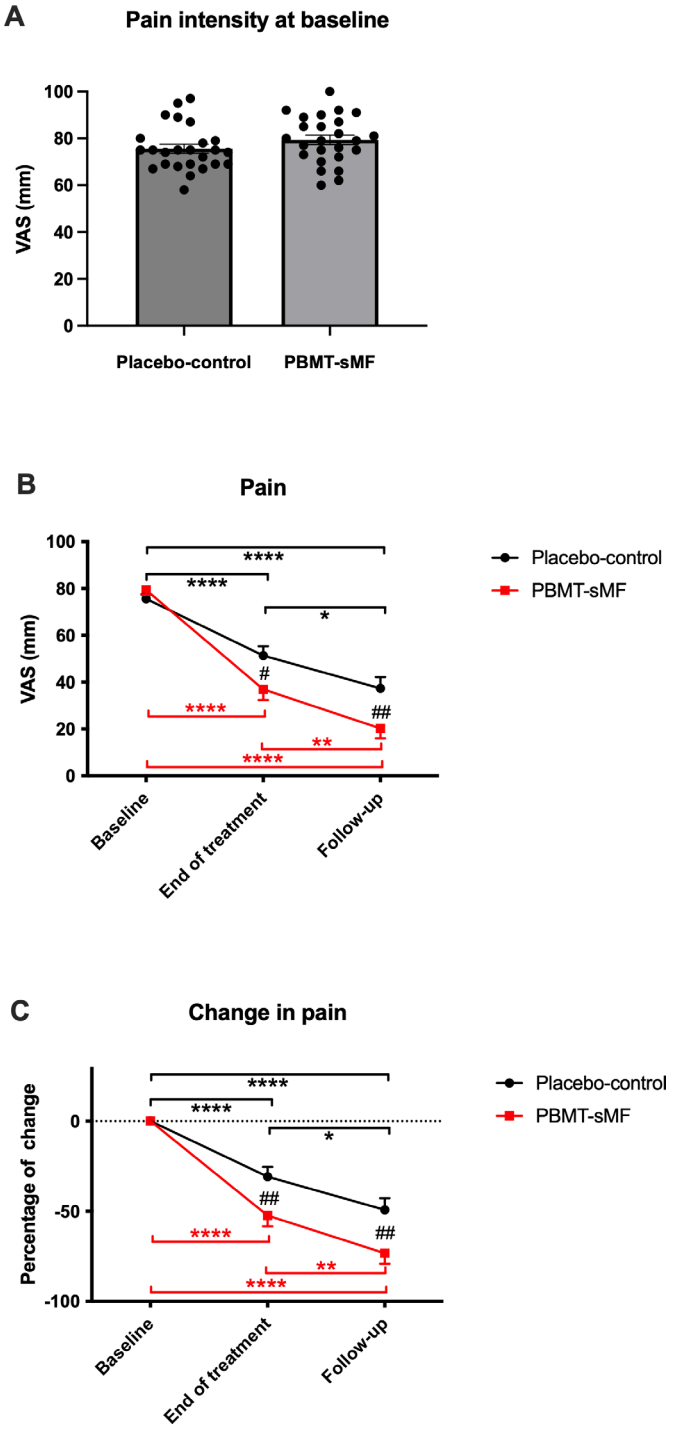


Figure 2 Pain intensity measured by VAS. (A) Mean ± SEM shown as bar graphs with individual scatter points (each dot represents one participant). (B, C) Mean ± SEM shown as line graphs; symbols represent group means at each time point, with error bars for SEM. Statistical comparisons were performed using ANCOVA, adjusting for baseline values. PBMT-sMF vs Placebo: ##p<0.01, #p<0.05. Within-group comparisons (vs baseline and end of treatment): ****p<0.0001, **p<0.01, *p<0.05. ANCOVA: one-way analysis of covariance; PMBT-sMF: photobiomodulation therapy combined with static magnetic field; SEM: SE of mean; VAS: visual analogue scale.

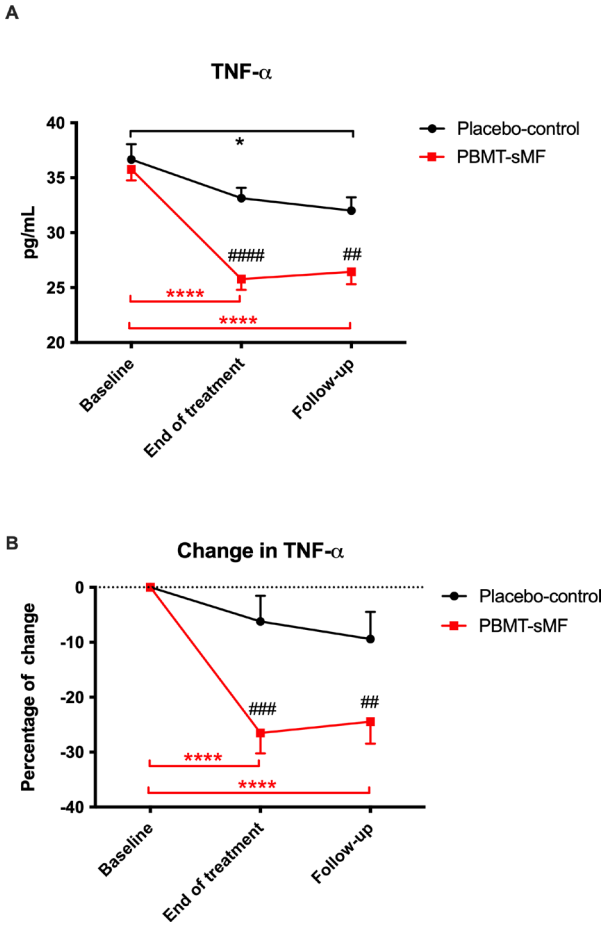


Figure 3 TNF-α levels measured from serum samples. (A, B) Mean ± SEM shown as line graphs; symbols represent group means at each time point, with error bars for SEM. Statistical comparisons were performed using ANCOVA, adjusting for baseline values. PBMT-sMF vs Placebo: ####p<0.0001, ###p<0.001, ##p<0.01. Within-group comparisons (vs baseline and end of treatment): ****p<0.0001, *p<0.05. ANCOVA: one-way analysis of covariance; PMBT-sMF: photobiomodulation therapy combined with static magnetic field; SEM: SE of mean; TNF-α: tumour necrosis factor-alpha; VAS: visual analogue scale.

Moreover, the aforementioned studies presented methodological limitations, including inadequate follow-up, lack of blinding,[27 28] no concealed allocation, no true randomisation and the absence of intention-to-treat analysis.[27]

Findings on PBMT's effectiveness in lateral epicondylitis can also vary within the same study. For instance, Emanet *et al*[18] evaluated short- and long-term (12 week) PBMT outcomes, noting no difference between groups initially, but a clear advantage for PBMT over placebo in functional measures at longer follow-up. These results partially mirror our findings, given that we did not observe between-group differences in functional parameters in the short term. On the other hand, we found PBMT to be superior to placebo in reducing pain intensity at both short-term and 1-month follow-up, consistent

Table 3 Mean and standard deviation (absolute values) for primary and secondary outcomes of the study (n=50)

Outcomes	PBMT-sMF (n=25)	Placebo (n=25)	Between-group difference
Pain intensity (VAS 0–100)			
Baseline	79.36 (10.19)	75.60 (9.64)	ns
End of treatment	36.92 (22.64)*	51.40 (19.84)*	†
Follow-up	20.28 (21.24)* ‡	37.40 (24.08)* §	¶
Forearm pain and disability (PRTEE 0–100)			
Total scores			
Baseline	58.26 (9.45)	59.28 (8.48)	ns
End of treatment	39.44 (7.53)	40.88 (8.61)	ns
Follow-up	29.60 (10.67)	33.34 (9.68)	ns
Grip strength (kgf)			
Affected/ treated limb			
Baseline	23.13 (8.16)	24.24 (12.04)	ns
End of treatment	25.07 (8.83)	26.37 (12.70)	ns
Follow-up	25.97 (7.53)	27.72 (12.60)	ns
Unaffected/ untreated limb			
Baseline	24.71 (7.77)	27.55 (12.46)	ns
End of treatment	25.45 (9.28)	26.67 (12.57)	ns
Follow-up	26.07 (8.72)	28.39 (11.23)	ns
TNF-α (pg/mL)			
Baseline	35.75 (4.99)	36.66 (6.96)	ns
End of treatment	25.77 (4.90)*	33.14 (4.71)	**
Follow-up	26.45 (5.69)*	32.00 (6.11)††	¶

PBMT-sMF vs Placebo: **p < 0.0001; ¶p < 0.01; †p < 0.05.
Difference over time within group: Compared to baseline: *p < 0.0001; ††p < 0.05.
Compared to the end of treatment: ‡p < 0.01; §p < 0.05.
Statistical analysis was performed using ANCOVA, adjusting for baseline values.
PBMT-sMF, photobiomodulation therapy combined with static magnetic field.ANCOVA, one-way analysis of covariance; ns, not significant; PRTEE, Patient-Rated Tennis Elbow Evaluation; TNF-α, tumour necrosis factor-alpha; VAS, visual analogue scale.

with other investigations.[19 20] Those studies additionally reported better outcomes with PBMT than placebo for range of motion, free-weight elevation,[19] grip strength[19 20] and mechanical pain threshold.[20] However, unlike our trial, they incorporated an exercise regimen alongside PBMT, which may have contributed to enhanced functional improvements. Despite similarities in pain-related findings, variations in PBMT protocols—such as higher doses and lower treatment frequency in our study, as well as the use of multiple wavelengths—could explain some discrepancies in results.

Finally, a systematic review assessed the effects of PBMT on lateral epicondylitis, accounting for potential confounders and conducting subgroup analyses based on treatment parameters.[21] Their findings are consistent with ours in showing that PBMT can outperform control groups in reducing pain intensity, both immediately post-treatment and at follow-ups ranging from three to 8 weeks. However, they identified 904 nm as the most effective wavelength and did not investigate multiple-wavelength approaches, as used in our trial. Even so, our results align with the notion that combining more than 1 wavelength can be effective and, in certain instances, may further enhance PBMT's pain-relieving benefits.[29] On the other hand, that same review indicated PBMT was superior to placebo for improving grip strength, a finding we did not replicate. Of the 5 studies included in this analysis, 3 employed an exercise protocol alongside PBMT, potentially contributing to the more favourable functional outcomes observed in those investigations.

The analgesic effects of PBMT in musculoskeletal disorders are believed to stem from 2 well-established mechanisms: inhibition of neural activity, modulating peripheral nervous system signalling[30] and modulation of inflammatory processes via the regulation of key biomarkers such as prostaglandin E2 (PGE2), interleukin-1β (IL-1β) and TNF-α.[31–34] In the present study, we did not assess outcomes directly related to neural function. However, our findings showed that PBMT-sMF significantly reduced TNF-α levels, suggesting that the anti-inflammatory pathway was the primary mechanism contributing to the observed analgesic effects in patients with lateral epicondylitis. TNF-α is a key pro-inflammatory cytokine involved in triggering and sustaining the inflammatory response.[35] It plays a role in pain mechanisms by sensitising nociceptors, increasing vascular permeability and stimulating the production of other mediators such as IL-1β and PGE2.[36] In conditions like lateral epicondylitis and other tendinopathies, higher levels of inflammatory cells have been associated with greater symptom severity and tissue degeneration.[37] Thus, the reduction in TNF-α observed after PBMT-sMF may reflect a meaningful anti-inflammatory effect, contributing to lower peripheral nociceptive input and resulting in pain relief.

These findings align with previous studies using similar technology—a combination of 3 wavelengths—to reduce

Table 4 Proportion of co-interventions in both treatment groups (n=50)

	Other co-interventions			Medication		
Time points	PBMT-sMF (n=25)	Placebo (n=25)	p-value	PBMT-sMF (n=25)	Placebo (n=25)	p-value
Stabilisation	1 (4)	2 (8)	>0.9999	4 (16)	1 (4)	0.3487
Procedure	2 (8)	2 (8)	>0.9999	3 (12)	3 (12)	>0.9999
Post-procedure	2 (8)	0 (0)	0.4898	1 (4)	4 (16)	0.3487

Categorical variables are expressed as number (%).
PBMT-sMF, photobiomodulation therapy combined with static magnetic field.

postoperative pain following hip arthroplasty.[33] Evidence indicates that a single wavelength may not sustain biomodulatory effects throughout a temporal window of 5 min to 24 hours. In contrast, the combined use of 3 distinct wavelengths has been shown to elicit biomodulatory responses at different time points within this window. Therefore, the simultaneous application of 3 wavelengths may ensure continuous biomodulation throughout the entire temporal range, as each wavelength activates biological effects at specific phases.[29] In contrast, we did not observe significant improvements in functional outcomes. This suggests that PBMT-sMF, when used alone and with the parameters tested, may not be sufficient to enhance muscle strength or physical function, despite its efficacy in reducing pain. This result was partially anticipated, given that no adjunct interventions—such as therapeutic exercise—were included to target strength or functional improvement. The intervention in this study was specifically focused on pain management.

No significant adverse effects were observed, consistent with previous findings,[38] supporting the safety of PBMT-sMF. Additionally, the proportion of responders in the PBMT-sMF group was substantially higher than in the placebo group (72% vs 40%), reinforcing the clinical relevance of this intervention. This between-group difference is both statistically significant and clinically meaningful, as it exceeds the commonly accepted threshold of a 30% reduction in pain intensity for meaningful improvement in pain trials.[25] In our study, individual treatment success was defined as a≥30% reduction in pain intensity from baseline. This approach is supported by the IMMPACT recommendations, which identify a 30% reduction as indicative of a moderate clinically important improvement in chronic pain populations.[25] This criterion has been widely adopted in randomised controlled trials and enables comparisons across studies with different baseline pain values, providing a more standardised interpretation of treatment response. Although Niemiec *et al*[39] proposed a fixed minimal clinically important difference of 15 mm on the VAS for patients with lateral epicondylitis, we opted for a percentage-based threshold, which accounts for interindividual variability and aligns with established methodological standards in pain research. Moreover, a 30% reduction generally exceeds or corresponds to the 15 mm threshold in patients with moderate-to-severe baseline pain, thus preserving the clinical significance of the response. Taken together, these findings support the use of PBMT-sMF, with the parameters applied in this study, as a safe and effective modality for reducing pain in individuals with lateral epicondylitis.

One limitation of this trial was that the effects of PBMT-sMF were evaluated only in the short term. In contrast, this trial demonstrated several methodological strengths. It was a multicentre study and employed rigorous blinding procedures for assessors, therapists and patients. The sample size was calculated to ensure sufficient power to detect meaningful differences between the interventions. In addition, the study was prospectively registered, adhered to the protocol without violations, and incorporated true randomisation, allocation concealment and intention-to-treat analysis. Notably, no participants were lost to follow-up. The inclusion of a placebo group helped control potential confounders such as the placebo effect, regression to the mean, therapist bias and co-interventions, which were monitored in both groups.

Future studies investigating the medium- and long-term effects of PBMT-sMF in patients with lateral epicondylitis are warranted. Furthermore, research comparing PBMT-sMF with other treatment options for lateral epicondylitis is necessary. Additional investigations should also assess the potential benefits of PBMT-sMF as an adjunct to exercise therapy aimed at improving functional outcomes. Finally, examining the effects of PBMT-sMF on other inflammatory markers in patients with epicondylitis is essential for confirming the underlying mechanisms by which PBMT-sMF exerts its effects in these patients.

CONCLUSIONS

The findings indicate that PBMT-sMF, administered with the tested parameters and treatment frequency, provides a significant benefit in reducing both pain intensity and TNF-α levels in patients with lateral epicondylitis.

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